The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Chapter 09 Stage 9 of 14 Extreme concentration · Hard to substitute

Memory and HBM

SK hynix, Samsung and Micron make nearly all high bandwidth memory (HBM), the stacked memory beside AI chips. SK hynix took 63.2 percent of HBM revenue in 2025.

Specimen 09 · Memory and HBM

A twelve-high HBM stack pulled apart: a logic die at the base, shown in copper, and twelve memory dies above it. The finished part is 11 by 11 mm and 0.72 mm tall. Each memory die is ground to about 30 micrometers, thinner than a human hair. The footprint is to scale, but the heights are exaggerated about fifteen times.

In plain terms

High bandwidth memory, or HBM, is where an AI chip keeps the numbers it is working on, and it is built to hand them over fast. A memory maker etches tiny holes into each memory wafer, fills them with copper and grinds the wafer until the copper shows through its back. It tests each of the wafer's chips and stacks up to sixteen good ones into a cube, with every layer wired to the ones below. The cube then sits beside the processor, the chip that does the calculating. Stacking is the hard part because each chip is ground thinner than a human hair and the finished cube has to stay within a height limit of less than a millimeter. One chip that fails after stacking can scrap the whole cube. Only SK hynix, Samsung and Micron make HBM in volume. One US export rule covers HBM from all three, including the two based in South Korea.

In short

US-based Micron went from 5.8 percent of HBM revenue in 2024 to 23.1 percent in the first quarter of 2026, according to IDC figures in SK hynix's US prospectus[1]. Over the same period, Samsung's share fell from 37.8 percent to 20.5 percent[1].

Since December 2024, a US license has been required, unless an exception applies, to export to China any HBM moving more than 2 GB per second per square millimeter of its area[2]. The Institute for Progress

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

reported in October 2025 that China's CXMT planned to produce HBM3 in 2026[3]. Even so, working from TrendForce's estimate of world HBM shipments, the institute projected that producers outside China would make 70 times as much HBM as China in 2026[3]. In June 2026 SemiAnalysis judged that CXMT was still struggling to stabilize supply of eight-high HBM3[4].

An AI accelerator moves numbers from memory to its arithmetic units and back. The US export control office wrote in December 2024 that a processor cannot reach its full capabilities unless its memory capacity and bandwidth rise with its speed[2].

Chokepoint card Memory and HBM, stage 9 of 14

Concentration	Substitutability	Price or market size
Extreme	**Hard**	**About $35B**
Three or fewer firms could supply this stage today, or one firm holds more than half of it.	Three firms took nearly all HBM revenue in 2025[1].	HBM market in 2025, Micron's December 2025 estimate[5]

Who leads	KR	**SK hynix**	63.2% of HBM revenue in 2025[1]
	KR	**Samsung**	19.3% of HBM revenue in 2025[1]
	US	**Micron**	17.4% of HBM revenue in 2025[1]
Where it is made	KR	**South Korea**	SK hynix in Cheongju and Icheon, Samsung in Pyeongtaek and Hwaseong
	US	**United States**	Micron headquarters and R&D, and an SK hynix packaging plant under construction in Indiana
	TW	**Taiwan**	Micron DRAM and HBM production
	JP	**Japan**	Micron Hiroshima, 1-gamma DRAM
	SG	**Singapore**	Micron HBM advanced packaging plant, adding capacity from 2027
Why substitution is slow	SK hynix says only a few firms have the expertise and resources to design, make and package memory chips[1]. It says HBM also needs advanced processes such as packaging with through-silicon vias (TSVs), the copper-filled holes that link the layers, and must pass customers' rigorous testing and approval[1].		

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

3 Should the US Sell Blackwell Chips to China? Institute for Progress, 25 October 2025. ifp.org

4 China's CXMT Is Set to Challenge DRAM Incumbents. SemiAnalysis, 23 June 2026. newsletter.semianalysis.com

5 Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology, 17 December 2025. s25.q4cdn.com

Where China stands

The Institute for Progress estimated in October 2025 that China would make virtually no HBM that year[3]. Huawei said in September 2025 that its Ascend 950 chips would use two HBM designs of its own, without saying who would manufacture the memory[6].

Where the US stands

Micron is the only American HBM maker[1]. As of October 2026, License Exception HBM lets US and allied exporters ship HBM below 3.3 GB per second per square millimeter without a license, including to China when the chip designer buys it directly and it goes straight to the packaging site[7].

01

How it works

Each layer in the cube is a DRAM chip, the ordinary kind of computer memory. Each layer is ground thin and bonded to the one below through tiny metal bumps called microbumps. The whole stack sits on a base die, a chip that manages it. The copper-filled holes are called through-silicon vias, or TSVs. They let a signal reach any layer without a wire running around the outside.

An HBM3E cube has a 1,024-bit interface, meaning it exchanges data with the processor over 1,024 wires at once[8]. That is sixteen times wider than a standard DDR5 memory module, the memory stick in a PC[8]. HBM4 doubles that to 2,048 wires[9]. Ordinary memory cannot be made that wide, because the pins would not fit on a circuit board. The stack can, since its wires run down through the TSVs and across a silicon interposer to the processor beside it, with no board in the way[8]. The interposer is a slice of silicon carrying fine wiring, set under both the memory and the processor (see Advanced packaging).

A chip needs the width most when it answers a prompt. Training reuses each stored number across a huge multiplication, so a chip can fetch little and compute a lot. But to generate one word of an answer, the chip reads the whole model from memory, plus the whole record of the conversation so far. Then it does a little arithmetic and waits for the next read. That record grows with the length of the conversation and the number of users served at once. So memory size and speed usually set how many people one chip can serve.

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

3 Should the US Sell Blackwell Chips to China? Institute for Progress, 25 October 2025. ifp.org

6 Groundbreaking SuperPoD Interconnect: Leading a New Paradigm for AI... Huawei, 18 September 2025. huawei.com

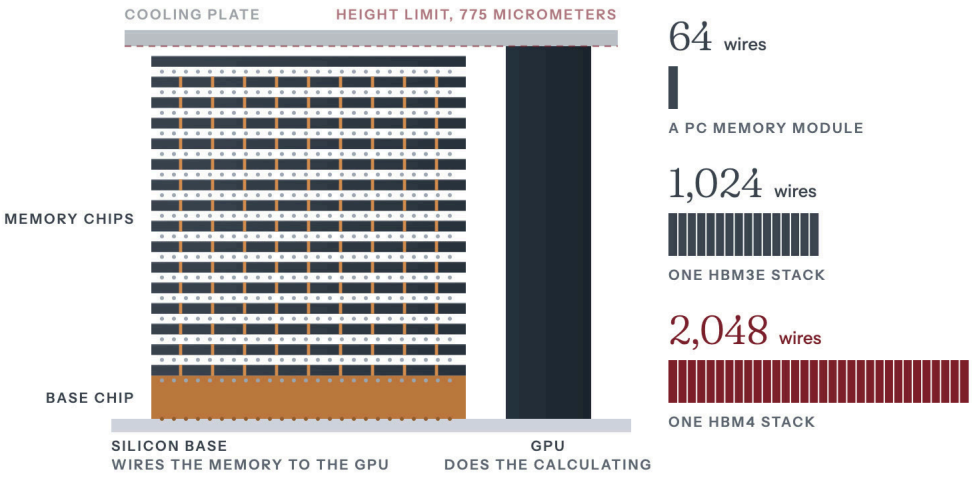
7 15 CFR 740.25 -- License Exception High Bandwidth Memory (HBM). Electronic Code of Federal Regulations, 5 December 2024. ecfr.gov

8 High-bandwidth memory (HBM). Micron Technology, accessed 9 September 2026. micron.com

9 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard... JEDEC Solid State Technology Association, 16 April 2025. jedec.org

Figure 09.1 · Explainer, 21 seconds

Inside a memory stack



1. **Base chip.** A memory stack sits beside the GPU on a shared silicon base. It starts with a base chip, which controls the memory chips above it.
2. **Thin.** Each memory chip is ground to about 30 micrometers, thinner than a human hair.
3. **Stack.** Twelve chips are joined by dots of solder, and copper-filled holes run straight down through all but the top one.
4. **Height.** The stack has to fit under the cooling plate. Twelve chips reach 0.72 mm, the old limit. HBM4 raised the limit to 775 micrometers and fits up to sixteen chips.
5. **Width.** An HBM3E stack talks to the processor over 1,024 wires at once, sixteen times as many as a PC memory module. HBM4 doubles that to 2,048.

Note Simplified cross-section. Heights follow the HBM height limits of 720 and 775 micrometers and are drawn to one vertical scale. Widths are not to scale.

02

Variants and trade-offs

Generation	Interface	Bandwidth per stack	Capacity per stack
HBM3E	1,024-bit, 16 channels (32 pseudo-channels)	over 1.2 TB/s	24 GB (8 chips), 36 GB (12 chips)
HBM4	2,048-bit, 32 channels	over 2.0 TB/s, up to 3.3 TB/s	up to 64 GB (16 chips, 32 Gb dies)

HBM3E

Micron's HBM3E comes in two sizes, 24 GB with eight chips stacked and 36 GB with twelve[10].

[10] Micron, HBM3E product page. Micron Technology, accessed 9 September 2026. micron.com

HBM4

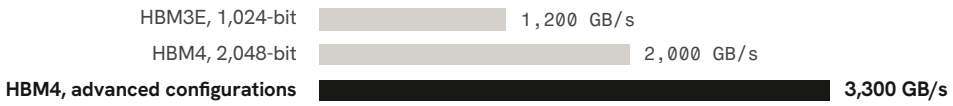
All three HBM makers run their HBM4 faster than the 8 Gb/s per pin in the industry standard. The standards body JEDEC published JESD270-4 in April 2025, setting the 2,048-bit interface and 2 TB/s per stack at 8 Gb/s per pin[9]. SK hynix says it was first to complete HBM4 development, in September 2025, at over 10 Gb/s per pin[11]. Samsung says it began mass production and shipped the first commercial HBM4 in February 2026, at 11.7 Gb/s per pin, with a 4 nm logic base die[12]. Micron's parts run above 11 Gb/s for more than 2.8 TB/s per stack[13]. Micron says it began volume production of its 36 GB twelve-high HBM4 in fiscal 2026 and expects to ramp volume of HBM4E, the next generation, in calendar 2027[14].

Through HBM3E, SK hynix made its own base dies, the chips at the bottom of each stack[15]. For HBM4 it chose TSMC's advanced logic process, so the base die can hold more functions[15]. As a result, TSMC, which does not make HBM, makes the base die for SK hynix's HBM4.

Figure 09.2

As of April 2026, HBM4 moves 2,000 GB/s per stack, HBM3E 1,200

Bandwidth per HBM stack



Source Siemens EDA, HBM3E and HBM4 IC design guide, April 2026 [16]

03

How a stack is built

- **Etch and thin.** Etch vertical holes, the through-silicon vias, into a memory wafer, fill them with copper, then grind the wafer from behind until the copper comes through.
- **Stack and join.** Thermocompression bonding joins the chips through metal bumps under heat and pressure, and it comes in several types, such as TC-NCF and MR-MUF, short for mass reflow

9 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard... JEDEC Solid State Technology Association, 16 April 2025. jedec.org

11 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix, 12 September 2025. news.skhynix.com

12 Samsung Ships Industry-First Commercial HBM4 With Ultimate Performance for AI... Samsung Global Newsroom, 12 February 2026. news.samsung.com

13 Micron, HBM4 product page. Micron Technology, accessed 9 September 2026. micron.com

14 MICRON TECHNOLOGY INC, Form 10-K annual report for the period ended 2026-09-03... U.S. Securities and Exchange Commission, 9 October 2026. sec.gov

15 SK hynix Partners with TSMC to Strengthen HBM Technological Leadership. SK hynix, 19 April 2024. news.skhynix.com

16 HBM3e and HBM4: IC design guide for next-generation high bandwidth memory. Siemens Digital Industries Software, 24 April 2026. blogs.sw.siemens.com

molded underfill[17]. SK hynix uses its own version, Advanced MR-MUF, which stacks the chips, injects a liquid protective material between them and hardens it[11]. SK hynix says the process controls warping and dissipates heat well[11]. Samsung said in February 2024 that its twelve-high HBM3E uses TC-NCF, thermocompression with a thin non-conductive film between the chips[18]. Samsung expected TC-NCF to help limit warping as chips get thinner[18].

- **Stay under the height limit.** The package specification for HBM4 raised the height allowed for a finished cube from 720 micrometers to 775, still under a millimeter[17].

Hybrid bonding is the next step, but it is not ready. It would join the chips copper to copper and insulator to insulator, with no solder and no filler[17]. In chip packaging generally, microbumps sit about 20 micrometers apart, and hybrid bonds can sit less than 1 micrometer apart[17]. A tech note on SK hynix's newsroom says full-scale adoption in HBM is most likely at HBM4E or HBM5, where stacks are projected to pass twenty layers[17]. Taller stacks trap more heat, and removing it becomes a major challenge at 16 to 20 layers[17]. The same tech note says hybrid bonding manages that heat more efficiently[17].

04

Who makes it

Three firms took more than 90 percent of the revenue from ordinary computer memory, or DRAM, in the first quarter of 2026, according to IDC figures in SK hynix's US prospectus[1]. HBM comes from the same three, SK hynix, Samsung and Micron[1]. No other company has both leading-edge DRAM and a stacking line of its own. SK hynix says only a few firms have the expertise and resources to design, make and package memory chips[1]. It says HBM also needs advanced production processes, such as TSV packaging, and must pass the rigorous testing and approval processes its customers set[1].

SK hynix says it was the first company to mass-produce several generations of HBM, including HBM3 and HBM3E[1].

SK hynix and Micron are both adding HBM capacity. SK hynix broke ground in August 2026 on an advanced packaging plant in West Lafayette, Indiana, an investment of over $4 billion[19]. Micron is modernizing and expanding its DRAM and HBM production capacity

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

11 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix, 12 September 2025. news.skhynix.com

17 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

18 Samsung Develops Industry-First 36GB HBM3E 12H DRAM. Samsung Global Newsroom, 27 February 2024. news.samsung.com

19 SK hynix breaks ground on $4 billion advanced packaging production facility in... Purdue University, 27 August 2026. purdue.edu

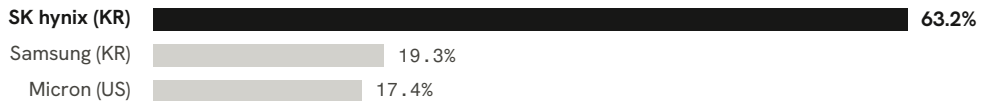
in Taiwan[14]. It has also broken ground on an HBM advanced packaging plant in Singapore to expand its packaging capacity from early 2027[14].

SK hynix led HBM revenue in 2024, in 2025 and in the first quarter of 2026, and in that quarter Micron passed Samsung for second place. In 2024 SK hynix took 56.4 percent, Samsung 37.8 and Micron 5.8[1]. In 2025 the three took 63.2, 19.3 and 17.4 percent[1]. In the first quarter of 2026 they stood at 56.4, 20.5 and 23.1 percent, which put SK hynix back at its 2024 share[1]. The split is public because SK hynix registered shares in the United States, and its US prospectus credits the research firm IDC[1].

Figure 09.3

SK hynix took 63.2 percent of global HBM revenue in 2025

Global HBM revenue by supplier, 2025



Source SK hynix Form 424B4 prospectus, 10 July 2026, on IDC data [1]

Figure 09.4

Micron sees HBM reaching $100 billion in 2028, from an estimated $35 billion

HBM market, Micron's estimate for 2025 and forecast for 2028



Source Micron prepared remarks, fiscal first quarter 2026, 17 December 2025 [5]

As HBM grows, memory makers have less factory capacity left for traditional DRAM. In December 2025 Micron put the HBM market at about $35 billion in 2025 and forecast that it would reach around $100 billion in 2028[5]. Micron says HBM needs more wafers than traditional DRAM, the kind used in PCs and phones, to make the same number of bits[14]. SK hynix says the industry's shift of production capacity to HBM has left traditional DRAM undersupplied[1]. The shortage, it says, helped drive a significant price recovery for that DRAM from the third quarter of 2025[1].

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

5 Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology, 17 December 2025. s25.q4cdn.com

14 MICRON TECHNOLOGY INC, Form 10-K annual report for the period ended 2026-09-03... U.S. Securities and Exchange Commission, 9 October 2026. sec.gov

05

The chokepoint

The leading accelerators made outside China use HBM from SK hynix, Samsung or Micron. Nvidia's GB300 NVL72 rack carries 20 TB of HBM3E at up to 576 TB/s[20]. Each Nvidia Rubin GPU carries up to 288 GB of HBM4 at up to 22 TB/s[21]. AMD's MI355X carries 288 GB of HBM3E at 8 TB/s[22].

A US export rule of December 2024 covers every HBM stack that was in production when it was published. The Bureau of Industry and Security (BIS), the US export control office, published the rule on 5 December 2024[2]. It added an export classification, ECCN 3A090.c, for "High bandwidth memory (HBM) having a 'memory bandwidth density' greater than 2 gigabytes per second per square millimeter"[2]. Its technical note defines that density as the memory bandwidth in gigabytes per second divided by the area of the package or stack in square millimeters[2]. The rule also covers stacks made outside the United States, through a foreign direct product rule, which applies US controls to goods made abroad with American technology[2]. BIS stated in the rule that all HBM stacks then in production exceeded that threshold[2].

BIS reviews license applications to ship controlled HBM to Macau or a Country Group D:5 destination such as China according to where the buyer is headquartered[23]. It presumes it will deny a license when the buyer or its ultimate parent is headquartered in one of those places, and presumes approval otherwise[23].

Two provisions narrow the control. License Exception HBM, at 15 CFR 740.25, applies to exporters headquartered in the United States or a Country Group A:5 ally[7]. It lets them ship HBM below 3.3 GB per second per square millimeter without a license, including to Macau and Country Group D:5 destinations such as China[7]. HBM sent to those destinations must be bought directly by the designer of the chip it will be packaged into[7]. Every shipment under the exception must go straight to the packaging site[7]. The rule also excludes co-packaged chips that combine HBM and logic and whose main function is processing, which other chip controls may cover instead[2].

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

7 15 CFR 740.25 -- License Exception High Bandwidth Memory (HBM). Electronic Code of Federal Regulations, 5 December 2024. ecfr.gov

20 NVIDIA GB300 NVL72. Nvidia, 23 July 2026. nvidia.com

21 Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia, 21 July 2026. developer.nvidia.com

22 AMD Instinct™ MI355X GPUs. AMD, accessed 9 September 2026. amd.com

23 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations, 23 July 2026. ecfr.gov

The think tank CSIS said in December 2024 that the rules still allowed HBM2, a generation introduced in 2016, to go to China, with limits on who could buy it and what for[24]. It also said Nvidia's H20, whose HBM3 is packaged with the processor, was then still exportable under the performance limits for AI chips[24].

China's strongest DRAM maker, CXMT, was still struggling to make HBM in steady supply by June 2026. In December 2024 CSIS called CXMT one of only two Chinese firms with a credible path to producing advanced HBM, though it did not name the other[24]. The Institute for Progress reported in October 2025 that CXMT was planning to produce HBM3 in 2026[3]. It also estimated that China would make virtually no HBM in 2025[3]. In June 2026 SemiAnalysis judged that CXMT was still struggling to stabilize supply of eight-high HBM3, with even greater challenges at twelve-high[4].

Huawei plans to use HBM of its own design in its Ascend 950 AI chips, but when it announced the designs in September 2025 it did not say who would make the memory[6]. The two designs are called HiBL 1.0 and HiZQ 2.0[6]. Huawei said the 950DT, due in the fourth quarter of 2026, would carry 144 GB of HiZQ 2.0 at 4 TB/s[6].

Four questions to answer from memory

1. What is HBM? see How it works
2. Why does an AI chip need such fast memory? see How it works
3. Who makes HBM? see Who makes it
4. How much HBM did China make in 2025? see The chokepoint

Four things to remember

- HBM is memory stacked into a cube beside the processor and wired to pass data fast.
- An AI chip reads its whole model from memory for each word, so memory speed limits it.
- SK hynix, Samsung and Micron make nearly all HBM.
- Since December 2024 US controls have covered the HBM stacks then in production, with two exceptions. The Institute for Progress estimated that China made virtually no HBM in 2025.

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3 Should the US Sell Blackwell Chips to China? Institute for Progress, 25 October 2025. ifp.org

4 China's CXMT Is Set to Challenge DRAM Incumbents. SemiAnalysis, 23 June 2026. newsletter.semianalysis.com

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Made by **Yashvardhan Sharma** with Claude. Corrections and questions: contact@chipsupplychain.org.

A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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